Metadata of the chapter that will be visualized in SpringerLink

Book Title	New Trends in Database and Information Systems
Series Title	
Chapter Title	Semantic Business Trajectories Modeling and Analysis
Copyright Year	2023
Copyright HolderName	The Author(s), under exclusive license to Springer Nature Switzerland AG

Corresponding Author	Family Name	**Arslan**
	Particle	
	Given Name	**Muhammad**
	Prefix	
	Suffix	
	Role	
	Division	
	Organization	Laboratoire Interdisciplinaire Carnot de Bourgogne (ICB)
	Address	Dijon, France
	Email	Muhammad.Arslan@u-bourgogne.fr
	ORCID	http://orcid.org/0000-0003-3682-7002
Author	Family Name	**Cruz**
	Particle	
	Given Name	**Christophe**
	Prefix	
	Suffix	
	Role	
	Division	
	Organization	Laboratoire Interdisciplinaire Carnot de Bourgogne (ICB)
	Address	Dijon, France
	Email	Christophe.Cruz@u-bourgogne.fr
	ORCID	http://orcid.org/0000-0002-5611-9479

Abstract

With the increasing availability of online news data through social media, there has been a growing focus on its potential as a source of business insights. However, interest in these insights has shifted towards more application-oriented analysis methods tailored to specific business purposes, leading to the emergence of semantically rich business trajectories. This article provides a definition of business trajectories and presents a method for constructing them using online news data. It also explores how trajectories can be enriched with semantic information to enable desired business interpretations and insights. Finally, the article discusses the potential of semantic business trajectories for environmental analysts in carrying out studies for various purposes.

Keywords (separated by '-')

Business trajectories - Environment - Knowledge management - NLP

Author Proof

Semantic Business Trajectories Modeling and Analysis

Muhammad Arslan(✉) and Christophe Cruz

Laboratoire Interdisciplinaire Carnot de Bourgogne (ICB), Dijon, France
{Muhammad.Arslan,Christophe.Cruz}@u-bourgogne.fr

Abstract. With the increasing availability of online news data through social media, there has been a growing focus on its potential as a source of business insights. However, interest in these insights has shifted towards more application-oriented analysis methods tailored to specific business purposes, leading to the emergence of semantically rich business trajectories. This article provides a definition of business trajectories and presents a method for constructing them using online news data. It also explores how trajectories can be enriched with semantic information to enable desired business interpretations and insights. Finally, the article discusses the potential of semantic business trajectories for environmental analysts in carrying out studies for various purposes.

Keywords: Business trajectories · Environment · Knowledge management · NLP

1 Introduction

Online news articles serve as a valuable resource for businesses seeking to identify opportunities across various industries [1]. These opportunities can range from a company advertising for labor to the sale of construction machinery, providing individuals or companies with the chance to capitalize on them [1]. These opportunities can be identified by grouping together multiple related key concepts, such as company, premises, project, and build, to form a business topic like "construction of premises."

By analyzing news articles using business topics, companies can classify business opportunities and gain valuable insights into the latest trends in their fields [2]. But business topics do not remain static over time. Instead, they follow a path of evolution or "trajectory" that can be influenced by a variety of external forces, such as changes in market trends, consumer behavior, technological advancements, or government regulations [2].

To obtain valuable insights from these trajectories, it is crucial to enhance them with supplementary information such as location and company details [2]. A semantic business trajectory (see Fig. 1) is a type of trajectory that has been enriched with annotations, which can provide additional contextual information. By representing this type of trajectory as a tuple (as mentioned below), businesses and researchers can gain a better understanding of the path of a particular business topic and how it evolves over time.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2023
A. Abelló et al. (Eds.): ADBIS 2023, CCIS 1850, pp. 1–11, 2023.
https://doi.org/10.1007/978-3-031-42941-5_33

Author Proof

(***trajectoryIdentification***, ***businessTopic***, ***trajectoryAnnotations***)
where:

trajectoryIdentification is the trajectory's unique identifier,
businessTopic defines the set of business concepts,
trajectoryAnnotations is the set of annotations (such as location, company name, time, etc.) associated to the trajectory.

Annotating business trajectories with location data can help identify geographic territories where business opportunities may be concentrated or emerging, while company information can reveal the source of the information and the specific organizations involved.

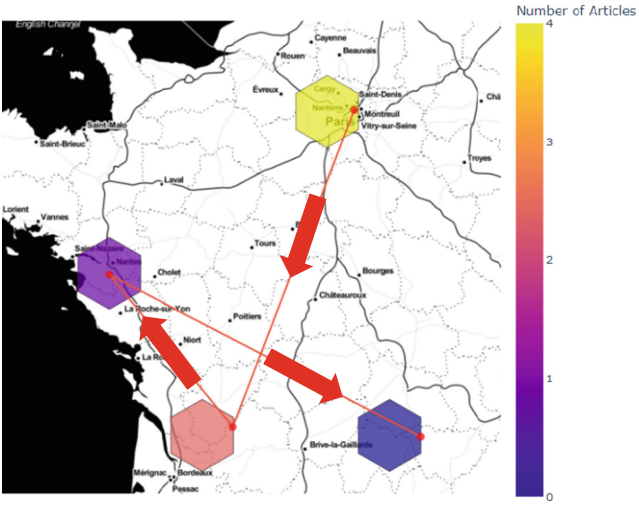

Fig. 1. A semantic business trajectory.

Natural Language Processing (NLP) techniques, such as topic modeling, can be used to identify key concepts from news articles, group them together to form business topics, and classify the articles under different topics. This process can be helpful in constructing business trajectories over time. Additionally, Named-Entity Recognition (NER) can be used to highlight important entities like locations and company names that are relevant to the topic, thereby providing semantic enrichment to the trajectories.

Semantic business trajectories have a wide range of applications beyond the business domain. They can also provide valuable insights into other areas, such as environmental science. By studying the impact of businesses across different geographical locations, we can gain a better understanding of their environmental impact. For example, we can analyze the carbon emissions resulting from various construction activities and identify potential ways to reduce them. This type of analysis is crucial for addressing climate change and other environmental challenges. Therefore, the use of semantic business trajectories can be an important tool for environmental scientists and researchers in a variety of fields.

Author Proof

The paper is structured as follows: Sect. 2 discusses the background of the research. Section 3 presents the proposed method for constructing semantic business trajectories. Section 4 discusses the presented method. The paper concludes Sect. 5 by discussing the potential directions for future research.

2 Background

2.1 From Business Concepts to Business Trajectories

Business trajectories are important in the field of Business Intelligence (BI) as they provide a way to track the evolution of business topics over time [1, 2]. However, before constructing business trajectories, it is essential to identify the relevant business concepts that form the basis of the trajectory. Several studies have focused on identifying and extracting business concepts from textual data, such as news articles, financial reports, and social media posts. One common approach is to use keyword extraction techniques to extract relevant terms and phrases that represent news articles.

In recent studies, various NLP techniques have been proposed to improve the extraction of relevant information from unstructured text data. For instance, in the study conducted by Huang et al. [3], an improved TextRank model is proposed for patent keywords extraction, which utilizes prior public knowledge such as public dictionary data to enhance the accuracy of the results. Another study by Wu et al. [4] proposes a rule-based approach for the mechanical-electrical-plumbing (MEP) NER and relationship extraction (RE), providing a more precise identification of relevant concepts in construction-related documents.

Additionally, Eroglu [5] proposed a text mining approach that employs the inverse document frequency method for trend tracking in scientific research. The author demonstrated the effectiveness of the proposed technique by applying it to a forest fire case study. Moreover, topic models such as BERTopic based on Bidirectional Encoder Representations from Transformers (BERT) can be used to identify key concepts as topics for text classification [6]. Traditional techniques, such as Latent Dirichlet Allocation (LDA), are limited in their ability to capture the evolution of topics over time. However, BERT-based dynamic topic modeling techniques overcome this limitation by modeling the evolution of topics and their representations over time. These studies illustrate the potential of NLP techniques in enhancing the extraction of valuable information from unstructured text data.

2.2 From Business Trajectories to Semantic Business Trajectories

The transformation of raw business trajectories into semantically rich trajectories is a challenge in this field. Several approaches and techniques [7–9] have been proposed that can help us to address this challenge and enrich business trajectories with semantic information. One of the primary techniques used to transform raw business trajectories into semantic trajectories is NER [10]. NER is a popular technique that identifies, and extracts named entities such as locations, organizations, and people from unstructured text data [9]. This technique has been used in various studies [10] to enrich business trajectories with location and company information.

Author Proof

Other techniques used to enrich business trajectories with semantic information include semantic web technologies such as Resource Description Framework (RDF) and Web Ontology Language (OWL), and knowledge graphs [11]. These techniques use ontologies and data models to represent the relationships between entities, providing a structured and semantically rich view of business trajectories. Several studies [12] have used these techniques to enrich business trajectories with additional information, such as spatio-temporal data and company details, which can provide valuable insights for businesses. By transforming raw business trajectories into semantically rich trajectories, businesses can gain a better understanding of their operations and how they are influenced by external factors, ultimately leading to more informed decision-making.

3 Semantic Business Trajectories

The literature review played a crucial role in identifying the competency questions (see Table 1) to model semantic business trajectories. Reviewing previous studies allowed us to identify the potential applications and use cases of semantic business trajectories, which formed the basis of the competency questions.

Table 1. Semantic business trajectories addressing competency questions.

No	Competency Questions	Required Information
1	What is the trend of business topics over time?	Business topics
2	What are the most common business topics overall, and how does their frequency change over time and across regions?	Business topics
3	How does the popularity of different business topics vary across regions?	Business topics and Locations
4	Which companies are most active in publishing news articles related to specific business topics in different regions?	Business topics, Locations, and Companies
5	What are the overall most important companies in terms of the frequency of news articles published about a business topic?	Business topics and Companies
6	Which companies are most important with respect to each region in terms of the frequency of news articles published about a business topic?	Business topics, Locations, and Companies

To address the competency questions using semantic business trajectories, the method is proposed. It mainly involves two stages:

1. Preprocessing the news data, and executing a topic model to identify relevant thematic concepts related to business trajectories using historical news article datasets, and

Author Proof

2. Semantically enriching the identified thematic topics using location and company information through NER.

To define the context of the study, the first step involves constructing a collection of relevant business concepts of interest. This is done by using a dataset of news articles that have been manually filtered to only include relevant articles. The news articles (8,300 in total), which were crawled from 95 different news sources such as lamontagne.fr, fusacq.com, businessimmo.com, sudouest.fr, actu.fr, ladepeche.fr, lavoixdunord.fr, etc., cover a time span of three months from Jan. 2023 to Mar. 2023 (see Fig. 2). To facilitate analysis, the news articles were translated from French into English using the Deep Translator Python library (https://pypi.org/project/deep-translator/) and underwent basic preprocessing steps such as stop-word removal, and filtering of special characters, weblinks, email addresses, and punctuations using the Natural Language Toolkit (NLTK) (https://www.nltk.org/).

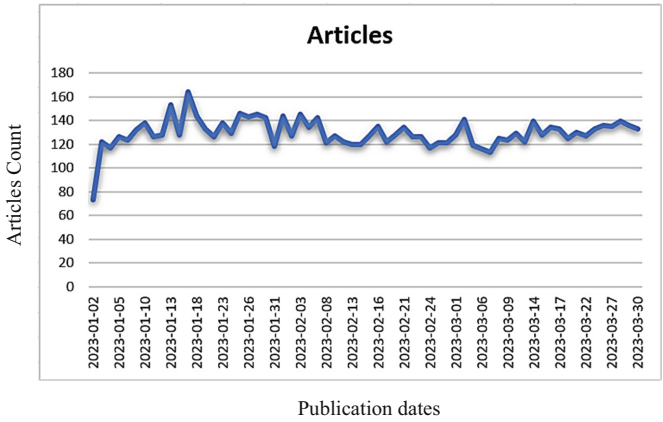

Fig. 2. Count of published news articles per date.

To identify the relevant thematic concepts from the text corpus, a topic model is applied. While there are many topic models available, BERTopic [6] is selected due to its efficiency and popularity in identifying underlying semantics in text and creating dense clusters for easily interpretable topics using transformers and c-TF-IDF (Term Frequency - Inverse Document Frequency) technique. The state-of-the-art BERTopic model [6] is trained on the news article corpus to extract topics (see Fig. 3). Each topic contains a group of related concepts. These concepts are represented in bar charts constructed from the c-TF-IDF scores for each topic representation. This allows for easy comparison of topics representations and their evolution over time (see Fig. 4). Once the topic model is trained, it can be used to predict the topics (as a set of related concepts) of new news articles. More details on BERTopic can be found in [6].

Figure 4 presents visualizations of business trajectories covering various business topics. It is evident from Fig. 4 that business topic 0 has the highest frequency and is the most dominant topic. For demonstration purposes, this topic's trajectory data has been used for further enrichment to gain more insights. For semantic enrichment, we have used

Author Proof

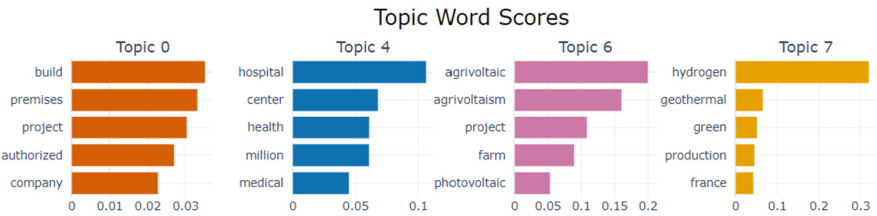

Fig. 3. Topic modeling.

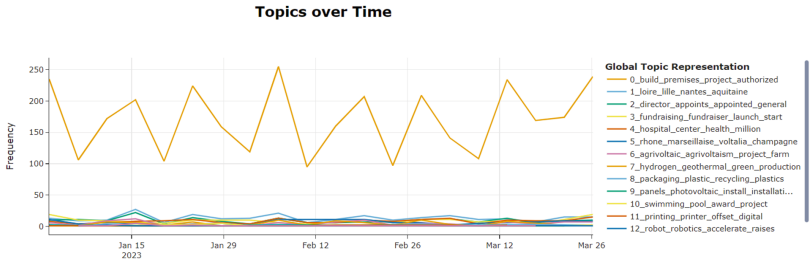

Fig. 4. Evolution of business trajectories over 3 months.

NER to identify and extract entities such as organizations and geographical locations from unstructured news data. There are various NER libraries available including SpaCy, Apache OpenNLP, and TensorFlow. Shelar et al. [13] found that SpaCy (https://spacy.io/) gives more accurate results as compared to Apache OpenNLP and TensorFlow to identify named entities in the text. Henceforth, SpaCy (https://demos.explosion.ai/) is used here (see Fig. 5).

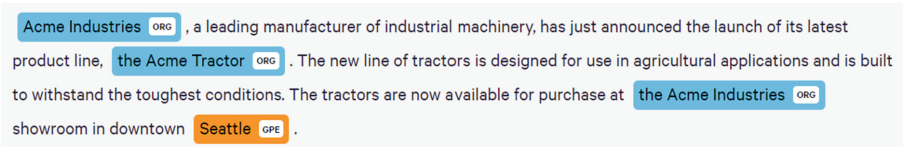

Fig. 5. NER.

By utilizing the location information extracted through NER, we can investigate the development of business topic 0 across various geographic regions (see Fig. 6). In Figs. 7 - 9, we present an analysis of the company information related to different geographic regions. Using NER, we extract company names from the news articles and associate them with their respective geographic locations. This allows us to track the presence and evolution of companies in different regions over time. By visualizing the figures, we can address the competency questions listed in Table 1. These visualizations allow us to gain insights into how the business topic of interest is evolving over time in different geographic regions and among different companies.

Author Proof

4 Discussion

The proposed method's novelty lies in the modeling of business news topics as semantic business trajectories, which provide an informative and engaging view of business topics in spatial, temporal, and contextual dimensions over time. Spatial information reveals whether a business topic is local or has a specific global political or economic facet based on the geographic territory. Temporal information reveals the importance and duration of the business topic. Contextual information of the trajectory shows how different geographic territories come up with the same business opportunities as competitors in the market. Also, company name information extraction will help us to combine more information (e.g., a SIREN number is a 9-digit identifier assigned to every registered business in France by the National Institute of Statistics and Economic Studies, and linked company contacts) related to the company from external data sources as shown in Fig. 10.

Modeled business trajectories can be used to address various environmental concerns [14–16]. For instance, issues related to construction activities (i.e., a business topic which is represented as a business trajectory), such as air quality waste management, and noise pollution. For example, to address air quality concerns, environmental analysts can use trajectory analysis to identify the geographic locations that are most affected by construction-related air pollution. This analysis can help environmental analysts develop targeted measures to reduce air pollution, such as endorsing cleaner construction materials, air filtration systems, or recommending construction activities during times of the day when air pollution levels are lower (Fig. 8).

AQ2

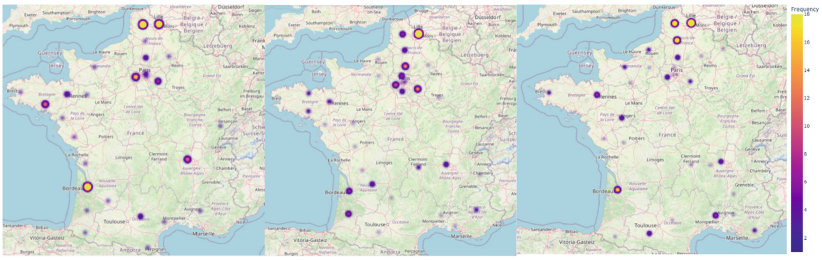

Fig. 6. Business topic evolution by publication count of news articles by regions.

Similarly, trajectory analysis can also be used to address waste management concerns. By analyzing the spatial and temporal distribution of construction activities identified through the published news articles (see Fig. 6), environmental analysts can identify areas where waste management is most needed and develop targeted strategies to reduce waste generation and ensure proper disposal of waste materials. Noise pollution can also be addressed using trajectory analysis. Environmental analysts can identify areas where noise levels are most significant due to increased construction activities (see Fig. 6) and recommend targeted strategies to reduce noise generation.

Environmental analysts could explore the use of semantic trajectory data from news articles to gain insights into how businesses are evolving in response to environmental

Author Proof

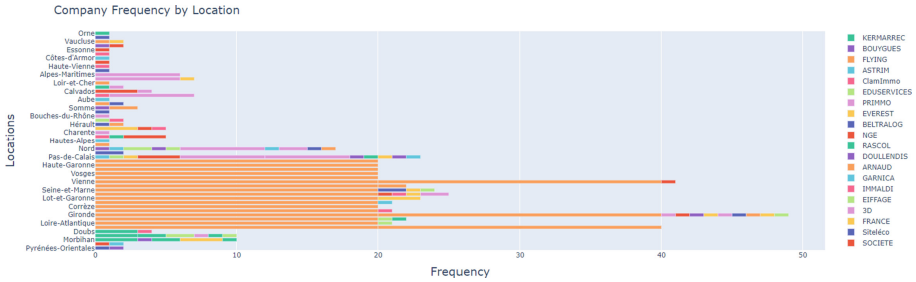

Fig. 7. Company and location information related to a business topic (Jan. 2023)

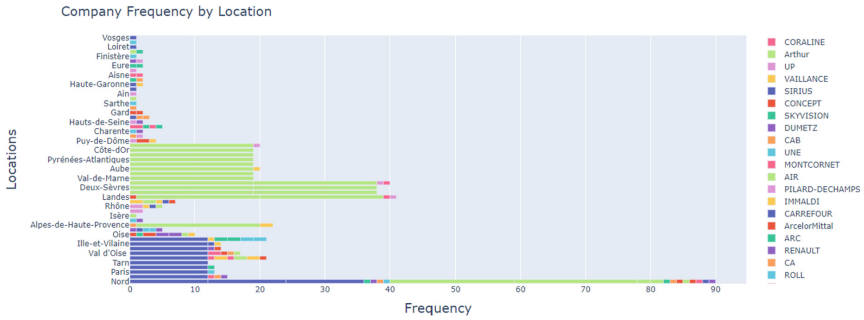

Fig. 8. Company and location information related to a business topic (Feb. 2023)

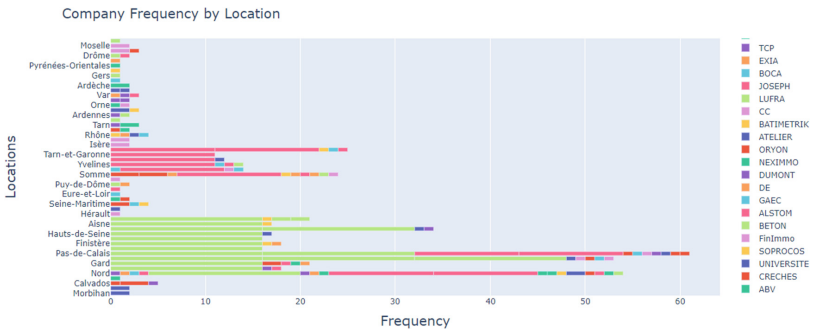

Fig. 9. Company and location information related to a business topic (Mar. 2023)

factors, such as temperature, in different geographical regions. This approach would provide a comprehensive understanding of the interplay between environmental factors and businesses over time. For instance, we visualize news articles related to topic 6 (“Photovoltaic”) from Fig. 3, along with temperature values of various geographical regions during a 1-month period (see Fig. 11). By doing so, we can identify which regions are publishing business opportunities related to “ Photovoltaic” over time, and how these regions are triggering opportunities in other regions (see Fig. 11). Furthermore,

Author Proof


```
[{'siren': '527561583',
  'zip': '49',
  'label': 'NATURAL',
  'alternateLabel': None,
  'contacts': ['Christopher Desailly']},
 {'siren': '000000526',
  'zip': None,
  'label': 'Barentz',
  'alternateLabel': None,
  'contacts': ['Hidde van der Wal']}]
```


Fig. 10. Company details extracted from an external source against company names.

this analysis can help analysts examine the relationship between temperature and regions, which is critical for generating business opportunities in the field of " Photovoltaic".

Despite numerous potential applications [17–19] for leveraging business datasets in environmental studies, these datasets are not currently being extensively integrated with environmental data. By utilizing knowledge management techniques (e.g., linked data concepts) to interlink data from business trajectories and environmental sources, we could gain deeper insights into the impact of business activities on the environment, identify patterns and trends, and devise strategies for sustainable development. This approach would not only improve environmental management but would also provide valuable knowledge assets for businesses, researchers, and policymakers to better understand the complex interplay between businesses and the environment.

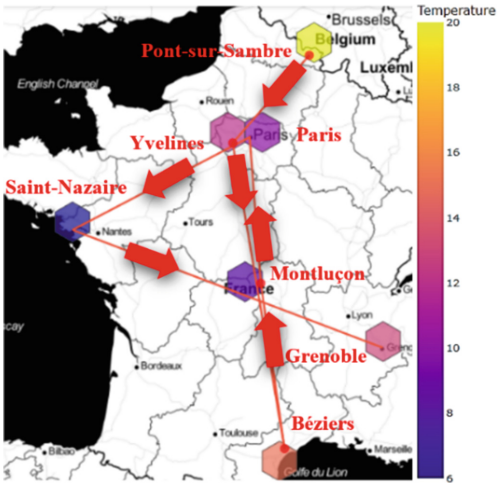

Fig. 11. Semantic business trajectory mapped with environmental data (i.e., temperature)

Author Proof

5 Conclusion

Online news articles offer a wealth of information about emerging business opportunities across various industries. By grouping related concepts into business topics, companies can gain valuable insights into the latest trends and changes in their fields. However, these business topics are not static and follow the path of "trajectory". To extract meaningful insights from these trajectories, it is essential to enrich them with additional information like location and company details. NLP techniques, including topic modeling and NER, are used for understanding business trajectories over time. Furthermore, semantic business trajectories have applications beyond the business domain and can offer valuable insights into areas like environmental science. By analyzing the impact of businesses across different geographical locations, researchers can gain a better understanding of their environmental impact and find ways to reduce carbon emissions and other environmental challenges. Therefore, the use of semantic business trajectories can be a crucial tool for researchers and scientists in various fields, including environmental science.

Acknowledgements. The authors thank the French government for the plan France Relance funding.

References

1. Arslan, M., Cruz, C.: Extracting business insights through dynamic topic modeling and NER. In: Proceedings of the 14th International Joint Conference on Knowledge Discovery (KEOD), pg. 215–222 (2022). https://doi.org/10.5220/0011552900003335
2. Arslan, M., Cruz, C.: Semantic taxonomy enrichment to improve business text classification for dynamic environments. In: INISTA, pp. 1–6. IEEE, Biarritz (2022)
3. Huang, Z., Xie, Z.: A patent keywords extraction method using TextRank model with prior public knowledge. Complex Intell. Syst. **8**(1), 1–12 (2022)
4. Wu, L.T., Lin, J.R., Leng, S., Li, J.L., Hu, Z.Z.: Rule-based information extraction for mechanical-electrical-plumbing-specific semantic web. Autom. Constr. **135**, 104108 (2022)
5. Eroglu, Y.: Text mining approach for trend tracking in scientific research: a case study on forest fire. Fire **6**(1), 33 (2023)
6. Grootendorst, M.: BERTopic: Neural topic modeling with a class-based TF-IDF procedure. arXiv preprint arXiv:2203.05794 (2022)
7. Solomon, Z., Ginzburg, K., Ohry, A., Mikulincer, M.: Overwhelmed by the news: a longitudinal study of prior trauma, posttraumatic stress disorder trajectories, and news watching during the COVID-19 pandemic. Soc. Sci. Med. **278**, 113956 (2021)
8. Zhang, Y., Zhang, H: FinBERT–MRC: financial named entity recognition using BERT under the machine reading comprehension paradigm. Neural Process. Lett. 1–21. (2023)
9. Novo, A.S., Gedikli, F.: Explaining BERT model decisions for near-duplicate news article detection based on named entity recognition. In: 17th International Conference on Semantic Computing (ICSC), pp. 278–281. IEEE (2023)
10. Arslan, M., Cruz, C.: Business insights using knowledge graphs by text analytics in dynamic environments. In: Proceedings of the 14th International Conference on Management of Digital EcoSystems, pp. 32–39 (2022). https://doi.org/10.1145/3508397.3564833
11. Memduhoglu, A., Basaraner, M.: An approach for multi-scale urban building data integration and enrichment through geometric matching and semantic web. Cartogr. Geogr. Inf. Sci. **49**(1), 1–17 (2022)

Author Proof

12. Arslan, M., Cruz, C.: Modeling virtual knowledge graphs using relevant news data by NLP methods for business analysis. In: 17th ICET, pp. 172–177. IEEE (2022). https://doi.org/10.1109/ICET56601.2022.10004674
13. Shelar, H., Kaur, G., Heda, N., Agrawal, P.: Named entity recognition approaches and their comparison for custom ner model. Sci. Technol. Libr. **39**(3), 324–337 (2020)
14. Lamba, P., Kaur, D.P., Raj, S., Sorout, J.: Recycling/reuse of plastic waste as construction material for sustainable development: a review. Environ. Sci. Pollut. Res. **29**(57), 86156–86179 (2022)
15. Atmaca, A., Atmaca, N.: Carbon footprint assessment of residential buildings, a review and a case study in Turkey. J. Clean. Product. 130691 (2022)
16. Szigeti, C., Major, Z., Szabó, D.R., Szennay, Á.: The ecological footprint of construction materials—a standardized approach from hungary. Resources **12**(1), 15 (2023)
17. Prakash, L.N.C.K., Suryanarayana, G., Jabbar, M.A.: Exploiting trajectory data to improve smart city services. Smart Urban Comput. Appl. **55** (2023)
18. Yu, D., Chen, Y.: The knowledge dissemination trajectory research of the carbon footprint domain: a main path analysis. Environ. Sci. Pollut. Res 1–18 (2022)
19. Arslan, M., Cruz, C.: Modeling semantic business trajectories of territories for multidisciplinary studies through controlled vocabularies. In: IEEE 39th International Conference on Data Engineering Workshops (ICDEW), pp. 170–177. IEEE (2023). https://doi.org/10.1109/ICDEW58674.2023.00032

Author Proof

Author Queries

Chapter 33

Query Refs.	Details Required	Author's response
AQ1	This is to inform you that corresponding author has been identified as per the information available in the Copyright form.	
AQ2	Please check and confirm if the inserted citation of Fig. 8 is correct. If not, please suggest an alternate citation.	